

① Modelling a collaboration Network

A mid-sized company analyzes weekly collaboration between 6 units.

Sales (S), Marketing (M), Logistics (L), IT (I)
Finance (F), HR (H)

⇒ The collaboration relations are:

(S, M), (S, L), (M, I), (L, I), (I, F), (F, H)

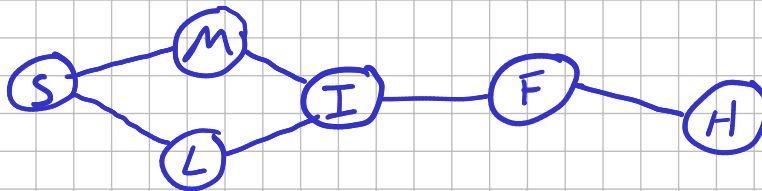
a) Define the node set V and edge set E

→ The nodes are the units in a company

$$V = \{S, M, L, I, F, H\}$$

$$E = \{(S, M), (S, L), (M, I), (L, I), (I, F), (F, H)\}$$

b) Draw the graph by hand.



c) Write the adjacency matrix in the order S, M, L, I, F, H

$$A = \begin{matrix} & \begin{matrix} S & M & L & I & F & H \end{matrix} \\ \begin{matrix} S \\ M \\ L \\ I \\ F \\ H \end{matrix} & \begin{bmatrix} 0 & 1 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 \end{bmatrix} \end{matrix}$$

* Since the graph is undirected, the adjacency matrix is symmetric!

d) Determine n , m , degree of each node, avg degree.

n = number of nodes = 6

m = number of edges = 6

$$k_s = 2$$

$$k_m = 2$$

$$k_l = 2$$

$$k_i = 3$$

$$k_F = 2$$

$$k_H = 1$$

Avg degree:

$$\langle k \rangle = \frac{2m}{n}$$

* Important!

In an undirected graph, each edge connects 2 nodes, therefore,

$$\frac{2m}{n}$$

Alternative Formula

$$\langle k \rangle = \frac{1}{n} \sum_{i=1}^n k_i$$

$$\langle k \rangle = \frac{1}{6} [2+2+2+3+2+1] = \frac{1}{6} [12] = 2$$

$$\langle k \rangle = \frac{2m}{n} = \frac{2 \cdot 6}{6} = 2$$

Laplacian:

$$Z = D - A$$

$$Z = \begin{array}{c} \begin{array}{c} S \quad M \quad L \quad I \quad F \quad H \\ \begin{array}{c|cccccc} S & 2 & 0 & 0 & 0 & 0 & 0 \\ M & 0 & 2 & 0 & 0 & 0 & 0 \\ L & 0 & 0 & 2 & 0 & 0 & 0 \\ I & 0 & 0 & 0 & 3 & 0 & 0 \\ F & 0 & 0 & 0 & 0 & 2 & 0 \\ H & 0 & 0 & 0 & 0 & 0 & 1 \end{array} \end{array} - \begin{array}{c} \begin{array}{c} S \quad M \quad L \quad I \quad F \quad H \\ \begin{array}{c|cccccc} S & 0 & 1 & 1 & 0 & 0 & 0 \\ M & 1 & 0 & 0 & 1 & 0 & 0 \\ L & 1 & 0 & 0 & 1 & 0 & 0 \\ I & 0 & 1 & 1 & 0 & 1 & 0 \\ F & 0 & 0 & 0 & 1 & 0 & 1 \\ H & 0 & 0 & 0 & 0 & 1 & 0 \end{array} \end{array} \end{array}$$

$$\begin{array}{c} \begin{array}{c} S \quad M \quad L \quad I \quad F \quad H \\ \begin{array}{c|cccccc} S & 2 & -1 & -1 & 0 & 0 & 0 \\ M & -1 & 2 & 0 & -1 & 0 & 0 \\ L & -1 & 0 & 2 & -1 & 0 & 0 \\ I & 0 & -1 & -1 & 3 & -1 & 0 \\ F & 0 & 0 & 0 & -1 & 2 & -1 \\ H & 0 & 0 & 0 & 0 & -1 & 1 \end{array} \end{array} \end{array}$$

③

Shortest paths in a product co-purchase network

An online retailer observes the following co-purchase network of six products:

$(P_1, P_2), (P_1, P_3), (P_2, P_3), (P_3, P_4), (P_4, P_5), (P_4, P_6), (P_5, P_6)$

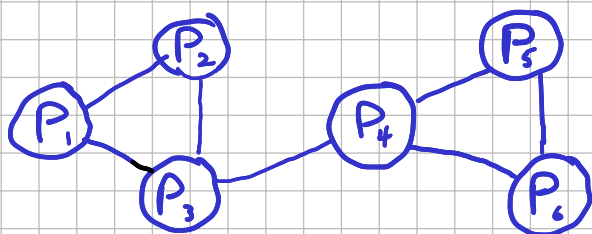
a) Draw the graph

→ we have 6 nodes:

$$V = \{P_1, P_2, P_3, P_4, P_5, P_6\}$$

→ we have the edges (relationships) of co-purchase:

$$E = \{\text{Same as above}\}$$



b) Compute the shortest path distances from P_1 to the other nodes

$$\begin{array}{lcl} d_{1,2} & = & 1 \\ d_{1,3} & = & 1 \\ d_{1,4} & = & 2 \\ d_{1,5} & = & 3 \\ d_{1,6} & = & 3 \end{array}$$

c) Compute the shortest path distances from P_4 to the other nodes.

$$\begin{array}{lcl} d_{4,1} & = & d_{4,3} + d_{3,1} = 2 \\ d_{4,2} & = & d_{4,3} + d_{3,2} = 2 \\ d_{4,3} & = & 1 \\ d_{4,5} & = & 1 \\ d_{4,6} & = & 1 \end{array}$$

d) Determine the diameter of the network

* Diameter is the longest shortest-path distance of all node pairs.

$$d_{1,6} = d_{1,5} = d_{2,6} = d_{2,5} = \textcircled{3}$$

e) Determine the APL

$$APL = \frac{1}{N(N-1)} \sum_{i,j=1}^N d_{ij}$$

$$APL = \frac{1}{6(6-1)} \cdot \left[(d_{12} + d_{13} + d_{14} + d_{15} + d_{16}) \right. \\ + (d_{21} + d_{23} + d_{24} + d_{25} + d_{26}) \\ + (d_{31} + d_{32} + d_{34} + d_{35} + d_{36}) \\ + (d_{41} + d_{42} + d_{43} + d_{45} + d_{46}) \\ + (d_{51} + d_{52} + d_{53} + d_{54} + d_{56}) \\ \left. + (d_{61} + d_{62} + d_{63} + d_{64} + d_{65}) \right]$$

$$= \frac{1}{30} (10 + 10 + 7 + 7 + 10 + 10)$$

$$= \frac{1}{30} \cdot 54 = \frac{27}{15} = \left(\frac{9}{5} \right) \quad \underline{\text{Average path Length}}$$

f) Explain what the APL means in the context of product discovery.

Low APL $\Rightarrow$ higher product exposure
(Fewer clicks to find related products)

High APL = siloed product
(customer is exposed to fewer related products)